

SC Operational Semantics:

Program transitions describe the steps in program executions, e.g. how a conditional (if) statement is executed.

Storage transitions describe how instructions interact with the storage (memory) system, e.g. how a memory read behaves.

- If the program takes a silent transition:

$$\frac{P, S \xrightarrow{\tau:\epsilon}_p P', S'}{P, S, M \rightarrow P', S', M}$$

- If both the program and storage systems take the same transition, then we can combine them:

$$\frac{P, S \xrightarrow{\tau:I}_p P', S' \quad M \xrightarrow{\tau:I}_m M'}{P, S, M \rightarrow P', S', M'}$$